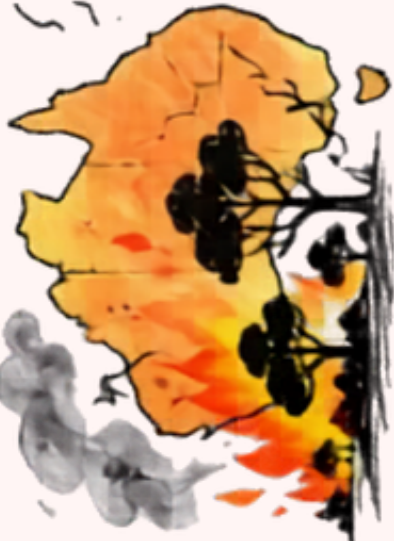




What Weather Conditions Drive Extreme Fire Risk in Australia?

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EMSC2010

A Bayesian Analysis of Meteorological Predictors of Extreme Fire-Risk Weather Days in Australia (2007-2017)



1. Research Question



“Which weather conditions are most strongly associated with periods of extreme fire risk in Australia?”

2. Data & Methods



Data

- WeatherAUS.csv (daily weather observations)
- BOM Sydney Observatory Hill rainfall data
- Period: 2007-2017
- Final dataset: 136,912 daily observations
- Variables: Max Temp (°C), Rainfall (mm), Humidity (%), ExtremeFire (binary).

ExtremeFire Definition (Proxy)



High Temperature (>90th percentile)



No Rainfall (0 mm)

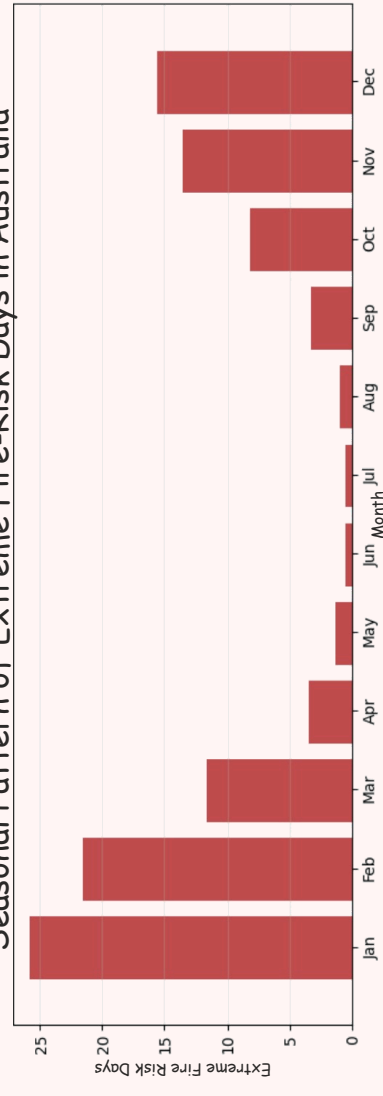
Methods



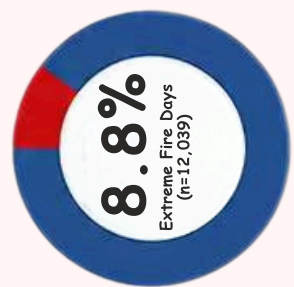
Exploratory Data Analysis Bayesian Two-Group Comparison Bayesian Logistic Regression Model Comparison (LOO-CV)

3. Key Results

Seasonal Pattern of Extreme Fire-Risk Days in Australia



Fire-risk days peak in January and February (Australian summer), confirming the expected seasonal pattern. Risk drops sharply in winter months (June-August). Bootstrap confidence intervals confirmed this pattern is statistically robust.

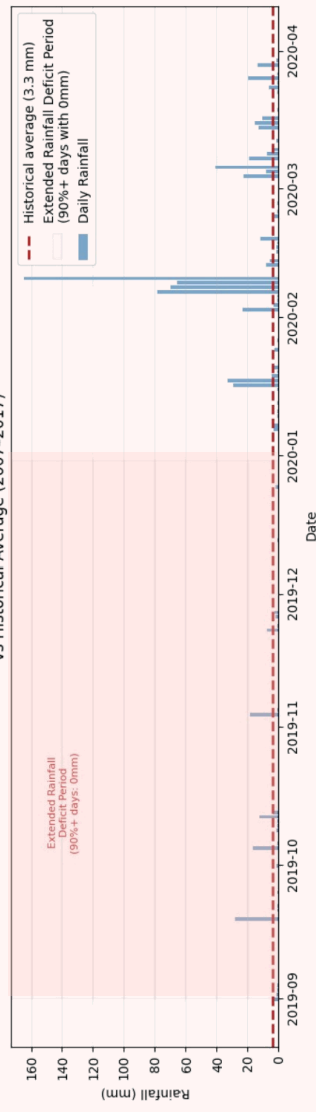


Normal Days (91.2%)
Extreme Fire (8.8%)

Posterior means and 95% HDI from the best-performing Bayesian logistic regression model (Temperature + Rainfall + Humidity). All HDIs exclude zero, confirming strong evidence for each predictor.

Variable	Posterior Mean	HDI 2.5%	HDI 97.5%	Direction
MaxTemp_scaled	5.166	4.522	5.884	Positive — higher temp increases risk
Rainfall_scaled	-2.356	-3.964	-0.675	Negative — more rain reduces risk
Humidity3pm_scaled	-0.412	-0.706	-0.133	Negative — higher humidity reduces risk

Daily Rainfall in Sydney During the 2019-2020 Black Summer vs Historical Average (2007-2017)



Daily rainfall in Sydney was near zero from September to December 2019 — well below the historical average of 3.3 mm/day. This extended drought created the conditions that drove the Black Summer fires. Heavy rainfall in early 2020 ended the season.

4. Main Findings



Temperature is the strongest driver of extreme fire-risk weather.



Low rainfall greatly increases the probability of extreme fire risk..



Higher humidity decreases the probability of extreme fire risk.



Bayesian methods allow us to quantify uncertainty rather than give a single value.

5. Limitations



- ExtremeFire is a simplified proxy, not actual fire occurrence.
- Important factors (wind, fuel load, vegetation, terrain) are not included.
- Dataset ends in 2017; external validation limited.

6. What I Learned

This project showed me that real-world data analysis is iterative, uncertain, and full of trade-offs. The gap between a clear research question and workable data required careful assumptions, adaptation, and justification at every step.

